



Technische Universität Berlin

Fakultät VII (Wirtschaft & Management)

Fachgebiet Wirtschafts- und Infrastrukturpolitik (WIP)

Energy Sector Modelling – EW MOD

The Effects of Regulatory Rules for Renewable Hydrogen

Examination of the Change in Energy System Cost

Authors:

Toyah Hohenstein (X) – toyah.hohenstein@tu-berlin.de

Leonard Schmitz (X) – leonard.schmitz@campus.tu-berlin.de

Paul Weishaupt (X) – weishaupt@campus.tu-berlin.de

Franz Wendel (X) – franz.wendel@campus.tu-berlin.de

Supervisor:

Dr. Konstantin Löffler

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Abstract

This paper investigates the impact of the European Union's regulatory framework for renewable hydrogen focusing on the requirements of additionality, temporal correlation, and geographical correlation on energy system costs and investment pathways. Using an extended energy system optimization model, four scenarios are analysed stepwise, ranging from an unconstrained baseline to full regulatory compliance. Results show that system costs increase by 21.1% under the full regulatory regime, with geographical correlation driving the largest share of additional costs. Compliance with the rules leads to a 29% increase in newly built renewable energy capacity and a 52% rise in electrolyser capacity, reflecting the need for greater flexibility in matching renewable generation with hydrogen production. The renewable share grows from 45% to 58.4%, while total emissions remain largely unaffected until 2040 and reach net-zero by 2050 through carbon capture and storage. Trade volumes decline as regional self-sufficiency gains importance, and unmet demand occurs only under the strictest scenario, signalling potential risks for system reliability. Overall, the results suggest that renewable hydrogen regulation effectively stimulates renewable deployment and supports decarbonisation, but at the cost of higher system expenditures and reduced flexibility. Simplifying assumptions in temporal and spatial modelling likely overestimate costs, highlighting the importance of future research with higher temporal resolution and more detailed spatial representation.

Contents

Statutory declaration	ii
Abstract	iii
Contents	iv
List of Figures	v
List of Tables.....	vi
List of Equations.....	vii
List of Abbreviations	viii
1 Introduction	1
2 Literature Review	2
2.1 EU-Regulation for Renewable Hydrogen.....	2
2.2 Related Work	3
3 Methodology and Data Assumptions	4
3.1 Initial Scenario	4
3.1.1 Cost of Unmet Demand.....	4
3.1.2 Additional Region	5
3.2 Regulatory Constraints	5
3.3 Assumptions and Limitations	8
3.3.1 Reduced Temporal Resolution and EU Regulations.....	8
3.3.2 Geographic Limitations	8
3.3.3 Simplified Resource and Infrastructure	8
4 Results and Discussion	9
4.1 System Costs	9
4.2 New Capacity	9
4.3 Total Capacity	11
4.4 Renewable Power Generation and Use for Hydrogen.....	12
4.5 Renewable Share	14
4.6 Emissions.....	14
4.7 Trade.....	15
4.8 Unmet Demand.....	15
5 Conclusion.....	16
References	vii
Appendix A.....	ix
Appendix B.....	xi

List of Figures

Figure 1: System costs of the scenarios [in million €]	9
Figure 2: Differences in RE and electrolyser capacities between scenarios [in GW].....	10
Figure 3: Development of total RE capacity across scenarios [in GW]	11
Figure 4: Development of total electrolyser capacity across scenarios [in GW].....	11
Figure 5: Changes in renewable power generation and use across scenarios [GWh]	12
Figure 6: Emissions in different scenarios from all technologies [in tons].....	14
Figure 7: Total trade differences	15

List of Tables

Table 1: Comparison of Literature Model Approaches and Results.....	3
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List of Equations

Equation 1: Objective Function (Default Model).....	4
Equation 2: Cost of Unmet Demand.....	4
Equation 3: Renewable Power for Hydrogen I	5
Equation 4: New Capacity Coupling I	6
Equation 5: Equal Power Use and Production I	6
Equation 6: Renewable Power for Hydrogen II	7
Equation 7: Equal Power Use and Production II	7
Equation 8: Renewable Power for Hydrogen III	7
Equation 9: New Capacity Coupling II.....	7
Equation 10: Equal Power Use and Production III	7

List of Abbreviations

Abbreviation	Meaning
A	Scenario: Additionality
AT	Scenario: Additionality + Temporal Correlation
ATG	Scenario: Additionality + Temporal Correlation + Geographical Correlation
CCS	Carbon Capture and Storage
EU	European Union
Hydro_ror	Hydropower from run of river
PV	photovoltaic
RE	Renewable Energy
RED III	EU Renewable Energy Directive
RE-Share	Renewable Share
VoLL	Value of Lost Load

1 Introduction

Since the adoption of the Paris Agreement, the global framework and ambition to limit global warming to well below 2 °C and ideally to 1.5 °C above pre-industrial levels have been firmly established. Achieving these targets requires rapid and substantial reductions in greenhouse gas emissions, with the ultimate goal of reaching net-zero emissions in the second half of this century. This implies balancing anthropogenic emissions with natural sinks and fundamentally transforming energy systems worldwide (Paris Agreement 2015).

For Europe, meeting such ambitious climate objectives necessitates a profound transition of its energy infrastructure. The German government considers green hydrogen a key technology that can play a central role in achieving deep decarbonization (Bundesministerium für Wirtschaft und Energie (BMWi) 2020). However, hydrogen is only considered truly sustainable if it is produced exclusively from renewable energy sources. This principle is reflected in Germany's National Hydrogen Strategy as well as in several EU-level regulatory initiatives. These include robust sustainability criteria, reliable certification schemes for renewable electricity and green hydrogen, and the development of harmonized European codes and standards to support the rapid scale-up of the hydrogen market (Bundesministerium für Wirtschaft und Energie (BMWi) 2020).

Given this context, the central research question of this paper is whether an independent and climate-neutral hydrogen supply is possible under new regulatory conditions and what additional costs arise from the requirements of additionality, temporal correlation and geographical correlation. The regulatory framework is a critical factor, as it shapes investment decisions, the expansion of new capacities, and ultimately the overall cost of the energy system. EU regulations require a strong link between hydrogen production and renewable electricity generation with respect to origin, time, and location requirements that could significantly reduce system flexibility.

The objective of this paper is to analyse the total system costs associated with implementing these regulatory requirements for renewable hydrogen and to assess how they influence the energy system's ability to ensure security of supply and decarbonization. Furthermore, the study investigates when and where additional capacities are needed and whether sufficient renewable electricity is available to meet future hydrogen demand.

The remainder of this paper is structured as follows: Section 2 provides an overview of the regulatory framework for renewable hydrogen under the EU Renewable Energy Directive (RED III) and reviews relevant literature on similar research questions. Section 3 introduces the energy system model used in this study and explains the methodological extensions implemented. Section 4 presents the results of the different implementation stages, focusing on the effects of additionality, temporal correlation, and geographical correlation. Section 5 discusses the findings and situates them within the broader literature, while Section 6 concludes by highlighting the main insights, limitations, and implications for future research.

2 Literature Review

This chapter has two aims. Section 2.1 sets the theoretical foundation for the model extension. Section 2.2 provides context and synthesizes related work to position the analysis of this study within current research.

2.1 EU-Regulation for Renewable Hydrogen

The following information on the regulations for renewable hydrogen is taken from the Delegated Regulation of the European Commission, published on 10 February 2023 (EU Regulation 2023/1184 2023).

Hydrogen is only considered renewable if it is produced using renewable electricity through electrolysis. Renewable energy (RE) can be generated from any renewable energy source except biomass (Article 2(3)). The installation generating electricity from RE sources must either be connected to the hydrogen production site via direct line (Article 3(a)) or under certain conditions, may also be connected to the grid (Article 3(c)). Electricity from the grid could be used e.g. if the share of RE in the bidding zone exceeds 90% (Article 4(1)). It may also be used if the emission intensity is below 18 g CO₂-eq/MJ and the temporal and geographical correlation requirements, as defined in the following sections, are fulfilled (Article 4(2)).

Furthermore, renewable electricity from the grid is allowed to be used if the additionality (Article 5), temporal correlation (Article 6) and geographical correlation (Article 7) requirements are met. This paper focuses exclusively on this type of renewable hydrogen.

The additionality states that hydrogen producers can only use renewable electricity from their own installations or through a power purchase agreement. The renewable electricity generation facilities must have been commissioned no earlier than 36 months before the first production of hydrogen (Article 5(a)). Furthermore, the facilities must not have received any form of operating or investment aid for their construction (Article 5(b)). During the transitional phase until 2038, the additionality requirement applies only to hydrogen production facilities commissioned from 2028 onwards (Article 11).

The temporal correlation provides that, until the end of 2029, electricity generation and hydrogen production must occur in the same month. From 2030 onwards, the rule becomes stricter, and the electricity must be produced in the same hour. Storage units can be used if they are located behind the same grid connection point as the electrolyzers and are charged and discharged within the same hour. Day-ahead market coupling is also permitted if the clearing price in the bidding zone is below 20 €/MWh or less than 0.36 times the price of a CO₂ allowance.

The geographical correlation requires that the electrolyser and the electricity generation facility must be in the same bidding zone at the time of commissioning (Subsection 1a). The facilities may also be located in different bidding zones if the zones are interconnected. In that case, the day-ahead market price in the generation zone must be at least as high as in the electrolyzers zone (Subsection 1b). Renewable electricity can also be used from interconnected offshore bidding zones (Subsection 1c). With regard to geographical correlation, each country may apply stricter criteria (Article 7(2)).

2.2 Related Work

To provide context for the following research, a literature review is conducted. Including criteria are publication dates between 2024 and 2025, the use of Operations Research methods to model the behaviour of the European Energy System holistically or by part and the implementation of constraints for renewable hydrogen according to EU Regulation 2023/1184 (2.1). Non-peer-reviewed publications are excluded. To establish a clear comparison, each study is analysed to extract both model and result related data.

Table 1 shows an overview of publications analysed. The models vary in scope and complexity, ranging from local single node (Brandt et al. 2024) to large multi-node models like EMPIRE and Balmorel (Hordvei et al. 2025; Kountouris et al. 2024; Petersen et al. 2025). Results indicate that an independent supply of climate-neutral hydrogen is technically possible but cost intensive.

Author (Year)	Model	Scope	Period	Key Result
Brandt et al. (2024)	custom linear model	EU (Single Locations, 1 node each)	2025-2030	+ 0.55 - 1.62 €/kg H ₂ production cost
Hordvei et al. (2025)	EMPIRE	EU (52 nodes)	2024-2048	+ 82 bln € (3,7%) system cost
Kountouris et al. (2024)	Balmorel	Europe + Neighbours (43 nodes)	2030-2050	+ 3 % system cost
Petersen et al. (2025)	EMPIRE	EU (52 nodes)	2024-2048	+ 80 bln € (3,6%) system cost
Zeyen et al. (2024)	PyPSA-Eur	GER/NL + Neighbours (nodes NA)	2025 and 2030	- 7 - 8 % temporality cost (flexibility)

Table 1: Comparison of Literature Model Approaches and Results

Considering the EU regulations for renewable hydrogen, system costs are likely to increase by 3 – 4%, which is around 80 billion € (Hordvei et al. 2025; Kountouris et al. 2024; Petersen et al. 2025). This is in line with an increase in production cost for renewable Hydrogen due to the EU regulations (Brandt et al. 2024). At the same time the cost impact of the temporal constraint can be reduced through grid flexibility. The introduction of storages and resulting temporal supply shift can reduce the temporal constraint cost impact by 7-8%, while ensuring the compliance with the remaining two renewable hydrogen constraints (Zeyen et al. 2024).

While recent studies use complex models of the European energy system, this research is positioned as a balanced approach between complexity and simplicity. High model complexity increases realism but often reduces transparency and makes it hard to isolate the impact of individual regulations (DeCarolis et al. 2017; Moglen et al. 2023). By using a simpler approach, this study may therefore lead to a clearer assessment of the EU regulation on system costs. Through this, this study complements existing work, while being focused on one topic.

3 Methodology and Data Assumptions

The goal of this term paper is to implement the regulatory rules for renewable hydrogen and to examine their impact on the system costs of the energy model. It also investigates whether a supply with renewable hydrogen is possible or if unmet demand occurs. The regulations are introduced step by step in the model, allowing for a detailed analysis of their individual effects. The analysis starts with the default model (initial scenario) without any renewable hydrogen rules. Three regulatory steps are then added sequentially: Additionality (scenario A), Temporal Correlation (scenario AT), and Geographic Correlation (scenario ATG), with each scenario including the restrictions of the previous one. Each step is evaluated in terms of changes in investment decisions (new and total capacity), emissions, renewable power generation and use for hydrogen, renewable share (RE-share), trade, and unmet demand. Every model run is saved as separate CSV files to ensure result reproducibility. The results are then processed in Tableau to compare changes across the scenarios and to analyse the combined effect when all regulations are applied. This method allows for a clear identification of the effects. The following sections describe the default model (3.1), the added constraints for Additionality, Temporal Correlation, and Geographic Correlation (3.2), and finally the key assumptions and limitations (3.3). In addition, Figure A 1 to Figure A 5, in Appendix A shows the complete Julia code that was developed.

3.1 Initial Scenario

The initial model acts as the reference and represents an energy system with no constraints regarding the production of renewable hydrogen. Hydrogen can be produced using any renewable and non-renewable energy capacity. The base model “*Model with hourly data & more technologies*” was used and extended by a realistic unmet demand penalty (3.1.1) and an additional region (3.1.2).

The model consists of 37 technologies (renewable and non-renewable), three storage technologies, nine fuels, six regions, interregional trade and costs. The objective is to minimize the system cost, which contains all costs described in the Equation 1 below.

$$\min \left(\sum \frac{\text{TotalCost}_{y,r,t}}{(1 + \text{DiscountRate})^{y-\min(\text{year})}} + \sum \frac{\text{TotalStorageCost}_{y,r,s}}{(1 + \text{DiscountRate})^{y-\min(\text{year})}} \right. \\ \left. + \sum \frac{\text{TradeCostFactor}_f * \text{TradeDistance}_{r,rr} * \text{Export}_{y,rr,r,h,f} * \text{YearlyDifferenceMultiplier}_y}{(1 + \text{DiscountRate})^{y-\min(\text{year})}} \right. \\ \left. - \sum \text{SalvageValue}_{y,r,t} + \sum \text{UnmetDemand}_{y,r,h,f} * \text{UnmetDemandPenalty}_{y,r} \right)$$

Equation 1: Objective Function (Default Model)

3.1.1 Cost of Unmet Demand

To have a feasible solution in hours where supply does not meet demand, the variable *UnmetDemand* is introduced to the supply side of the energy balance. To discourage the model to use this variable extensively, it is associated with cost and considered in the objective function as *CostofUnmetDemand*. To calculate the cost, *UnmetDemand* is multiplied with the penalty factor *UnmetDemandPenalty*, which is introduced as region and year dependent as described by Equation 2.

$$\text{CostofUnmetDemand}_{y,r,h,f} = \sum \text{UnmetDemand}_{y,r,h,f} * \text{UnmetDemandPenalty}_{y,r}$$

Equation 2: Cost of Unmet Demand

UnmetDemandPenalty is defined as the Value of Lost Load (VoLL). As input data, regional averages for households and industrial sectors are used as shown in Table B 1. The representation of economical differences through VoLL (Schröder and Kuckshinrichs 2015) is achieved, as they differ widely between regions. Still, due to different VoLL model approaches and high variance between sectors, these values underly uncertainty. In addition to the base VoLL in year 2020, positive VoLL growth until the year 2050 due to electrification (Broberg et al. 2021) and positive correlation with GDP per capita (van der Welle and van der Zwaan 2007) is assumed.

3.1.2 Additional Region

In addition to the existing regions, the model is extended southwards to have a more realistic model. Spain is added as an additional region. Spain related data was taken from „ew-mod25_model_final_hourly_ES“. In addition, *maximumtradecapacity.csv* (Table B 2) and *tradedistance.csv* was extended. It is assumed that spains whole trade is via land, with no other trade region connected by ports or terminals. Every region represents one bidding zone. Offshore is not considered independently but integrated into the respective regions.

3.2 Regulatory Constraints

In scenario A, the additionality requirement and the mandatory use of RE for hydrogen production via electrolysis are introduced. In this context, the CSV files *tagrenewable.csv* and *tagelectrolyzer.csv* were added. Both files consist of a single dimension and use 0 as the default value. In *tagrenewable.csv*, all renewable technologies eligible for renewable hydrogen production are marked with a value of 1. A total of five technologies is defined as renewable, offshore wind, onshore wind, photovoltaic (PV), hydro run-of-river and hydro from reservoir. The same principle applies to *tagelectrolyzer.csv*, where all electrolyser technologies are assigned a value of 1, while all others remain 0. In the model only alkaline electrolysis is included. The two CSV files are read into the model as the parameters *TagRenewable* and *TagElectrolyzer*. This approach allows additional renewable technologies or electrolyzers to be incorporated at any time without requiring changes to the code.

To ensure that hydrogen production is based exclusively on renewable electricity, the constraint *RenewablePowerForHydrogen* described in Equation 3 was implemented. The specification *fuels = "power"* ensures that the equation is formulated exclusively for electricity. The constraint stipulates that the total electricity consumption of all electrolyzers in the model is permitted only if it is covered by renewable energy generation.

$$\begin{aligned} & \forall y \in \text{year}, \quad \forall f \in \text{fuels}, \quad f = \text{power} \\ & \sum_{r \in R} \sum_{h \in H} \sum_{\substack{t \in T \\ \text{TagRenewable}=1}} \text{FuelProductionByTechnology}_{y,r,h,t,f} \geq \\ & \sum_{r \in R} \sum_{h \in H} \sum_{\substack{t \in T \\ \text{TagElectrolyzer}=1}} \text{FuelUseByTechnology}_{y,r,h,t,f} \end{aligned}$$

Equation 3: Renewable Power for Hydrogen I

To comply with the additionality requirement, two constraints were introduced. The constraint *NewCapacityCoupling* described in Equation 4 specifies that the expansion of electrolyzers must always linked to the parallel expansion of renewable electricity generation capacity. The new electrolyser capacities are multiplied by their electricity input demand to calculate the maximum electricity demand

of the newly built units. This demand may not exceed the annual electricity output provided by newly installed renewable capacity. In this way, the model ensures that additional electrolyzers can only be deployed if sufficient new renewable generation capacity is added in the same year.

$$\forall y \in \text{year}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{r \in \text{regions}} \sum_{\substack{t \in \text{technologies} \\ \text{TagElectrolyzer}=1}} \text{NewCapacity}_{y,r,t} * \text{InputRatio}_{t,f} \leq \sum_{r \in \text{regions}} \sum_{\substack{t \in \text{technologies} \\ \text{TagRenewable}=1}} \text{NewCapacity}_{y,r,t} * \text{OutputRatio}_{t,f}$$

Equation 4: New Capacity Coupling I

Since RE is not continuously available and subject to strong fluctuations, renewable generation capacities cannot be operated at full load throughout the year. It must therefore be ensured that the actual annual renewable electricity generation is sufficient to cover the demand of the newly installed electrolyzers. For this purpose, the constraint *EqualPowerUseAndProduction* described in Equation 5 is introduced. To calculate the available electricity, the newly installed renewable capacities are multiplied by the capacity factor, representing the share of capacity usable in each hour, and the output ratio. The electricity demand of the newly installed electrolyzers is represented by the actual electricity input of all electrolyzers (*FuelUseByTechnology*), minus the electricity demand of existing capacities. Existing capacities are defined as the accumulated capacity minus newly installed capacity. Their electricity demand is again calculated by multiplying with the input ratio and the (constant) capacity factor of electrolyzers. By multiplying both sides of the equation with $8760/n_{\text{hour}}$, the constraint is scaled to annual hours.

$$\forall y \in \text{year}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{r \in \text{regions}} \sum_{h \in \text{hour}} \sum_{\substack{t \in \text{technologies} \\ \text{TagRenewable}=1}} \text{NewCapacity}_{y,r,t} * \text{CapacityFactor}_{r,h,t} * \text{OutputRatio}_{t,f} * \frac{8760}{n_{\text{hour}}} \geq$$

$$\sum_{r \in \text{regions}} \sum_{h \in \text{hour}} \sum_{\substack{t \in \text{technologies} \\ \text{TagElectrolyzer}=1}} \text{FuelUseByTechnology}_{y,r,h,t,f} - \text{InputRatio}_{t,f} * \text{CapacityFactor}_{r,h,t} * \frac{8760}{n_{\text{hour}}}$$

$$(\text{AccumulatedCapacity}_{y,r,t} - \text{NewCapacity}_{y,r,t}) * \text{CapacityFactor}_{r,h,t} * \frac{8760}{n_{\text{hour}}}$$

Equation 5: Equal Power Use and Production I

For the introduction of temporal correlation (scenario AT) and geographic correlation (scenario ATG), no new constraints were developed, instead the three constraints from scenario A were extended. For this purpose, the indices of the constraints were adjusted. While in scenario A the constraints only had to hold on an annual basis, temporal correlation introduces the hourly index h in hour into the definition of the constraint. As a result, the equations are no longer aggregated across all hours but formulated for each individual hour. This requires that in every hour h the electricity generated from renewable capacities must at least cover the electricity demand of the newly installed electrolyser capacities in that hour. The constraint New Capacity Coupling (Equation 4) remains unchanged, as the hourly index has no effect on this condition. The other two constraints are modified as described in Equation 6 and Equation 7.

$$\forall y \in \text{year}, \quad \forall h \in \text{hour}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{r \in R} \sum_{\substack{t \in T \\ \text{TagRenewable}=1}} \text{FuelProductionByTechnology}_{y,r,h,t,f} \geq \sum_{r \in R} \sum_{\substack{t \in T \\ \text{TagElectrolyzer}=1}} \text{FuelUseByTechnology}_{y,r,h,t,f}$$

Equation 6: Renewable Power for Hydrogen II

$$\forall y \in \text{year}, \quad \forall h \in \text{hour}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{r \text{ in regions}} \sum_{\substack{t \text{ in technologies} \\ \text{TagRenewable}=1}} \text{NewCapacity}_{y,r,t} * \text{CapacityFactor}_{r,h,t} * \text{OutputRatio}_{t,f} * \frac{8760}{n_{\text{hour}}} \geq \sum_{r \text{ in regions}} \sum_{\substack{t \text{ in technologies} \\ \text{TagElectrolyzer}=1}} \text{FuelUseByTechnology}_{y,r,h,t,f} - \text{InputRatio}_{t,f} * (\text{AccumulatedCapacity}_{y,r,t} - \text{NewCapacity}_{y,r,t}) * \text{CapacityFactor}_{r,h,t} * \frac{8760}{n_{\text{hour}}}$$

Equation 7: Equal Power Use and Production II

Similarly, for geographic correlation the regional index r in region is implemented into the constraint definition. Instead of a systemwide aggregation, the condition is thus verified separately for each region. Accordingly, electrolyzers must cover their additional electricity demand with the output of newly installed renewable generation capacity in the same region. The constraints are modified as described by Equation 8, Equation 9 and Equation 10 below.

$$\forall y \in \text{year}, \quad \forall h \in \text{hour}, \quad \forall r \in \text{regions}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{\substack{t \in T \\ \text{TagRenewable}=1}} \text{FuelProductionByTechnology}_{y,r,h,t,f} \geq \sum_{\substack{t \in T \\ \text{TagElectrolyzer}=1}} \text{FuelUseByTechnology}_{y,r,h,t,f}$$

Equation 8: Renewable Power for Hydrogen III

$$\forall y \in \text{year}, \quad \forall r \in \text{regions}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{\substack{t \text{ in technologies} \\ \text{TagElectrolyzer}=1}} \text{NewCapacity}_{y,r,t} * \text{InputRatio}_{t,f} \leq \sum_{\substack{t \text{ in technologies} \\ \text{TagRenewable}=1}} \text{NewCapacity}_{y,r,t} * \text{OutputRatio}_{t,f}$$

Equation 9: New Capacity Coupling II

$$\forall y \in \text{year}, \quad h \in \text{hour}, \quad r \in \text{regions}, \quad \forall f \in \text{fuels}, \quad f = \text{power}$$

$$\sum_{\substack{t \text{ in technologies} \\ \text{TagRenewable}=1}} \text{NewCapacity}_{y,r,t} * \text{CapacityFactor}_{r,h,t} * \text{OutputRatio}_{t,f} * \frac{8760}{n_{\text{hour}}} \geq \sum_{\substack{t \text{ in technologies} \\ \text{TagElectrolyzer}=1}} \text{FuelUseByTechnology}_{y,r,h,t,f} - \text{InputRatio}_{t,f} * (\text{AccumulatedCapacity}_{y,r,t} - \text{NewCapacity}_{y,r,t}) * \text{CapacityFactor}_{r,h,t} * \frac{8760}{n_{\text{hour}}}$$

Equation 10: Equal Power Use and Production III

3.3 Assumptions and Limitations

The model presented in this paper is based on simplifying assumptions that are necessary to balance interpretability and complexity. However, these assumptions also lead to limitations regarding interpretation and transferability of the results to reality and must therefore be addressed.

3.3.1 Reduced Temporal Resolution and EU Regulations

The model consists of two timescales: One large timescale from 2020 to 2050 with 10-year steps for calculations between decades and a more granular scale for calculations within each year. The temporal resolution of the yearly scale is 120 hours. Regulatory requirements (2.1) are implemented from the start without a transitional period, with following exceptions: The “90% RE-share rule” for regions not to be affected by constraints is not considered. The “36-month rule” for additionality is not considered due to the temporal resolution of the 10-year time scale. The monthly aggregation option until 2029 is not considered due to the temporal resolution of the hourly timescale. While the lower resolution is computationally efficient, it is not sufficient to represent the temporal and geographical EU regulations which require at least hourly information: Renewable electricity and hydrogen must be produced in the same hour and storages must charge or discharge within that hour. In addition, cross zonal trade, supply and demand would be considered more realistically due to greater temporal detail in day-night cycles, daily variability, curtailment and fuel-flows. Modelling of even higher resolution, e.g. 15 min intervals to model day-ahead prices adequately, is not pursued to balance detail with the ability to solve the model with a reasonable number of computational resources. The described simplifications are likely to overestimate cost due to stricter assumptions and underestimation of grid flexibility.

3.3.2 Geographic Limitations

With the geographic constraint implemented, electrolyzers may only use electricity from generation plants within the same region. Bidding zones are not considered. This leads to not considering the allowed use of renewable power from neighbouring or offshore zones (2.1). Overall, geographic limitations lead to an underestimation of flexibility and the therefore an overestimation of costs.

3.3.3 Simplified Resource and Infrastructure

The model assumes unlimited availability and zero cost of water for electrolysis. This is an underestimation. Infrastructure like pipelines, storage systems and trade routes are simplified. The possibility to use the storage systems for electrolyzers in particular situations (2.1) is not considered. Spain trade routes are limited to pipelines for gas and hydrogen and power to France. As there are no hydrogen or gas terminals considered, France is Spain's only trade region. The energy system is isolated and no trade from outside is considered. These assumptions lead to overestimations due to lower flexibility. The VoLL is a region and year depended average value. It does not consider sector or fuel demand specific attributes. Depending on occurring unmet demand cost could be under or overestimated.

4 Results and Discussion

In the following chapter, the results of the optimisation are presented and subsequently analysed. First, the system costs (objective value) are examined, followed by a detailed analysis of the increase in renewable energy and electrolyser capacities, as well as the changes in total capacities resulting from the implementation of the renewable hydrogen regulations. This is followed by an assessment of emissions, hydrogen generation and use, and the changes in the electricity generation mix. Finally, the impacts on trade and unmet demand are analysed.

4.1 System Costs

The system costs for all four scenarios are shown in Figure 1. It can be observed that they increase with introduction of renewable hydrogen regulations depending on the scenario. The geographic correlation (scenario ATG) causes the largest rise at 18.46%, while additionality (scenario A) leads to cost increase of only 1.25% and temporality (scenario AT) raises system costs by just 0.02%. Overall, system costs increased by 21.11%, from 658,381 to 797,341 million €, when all three regulations are applied. This indicates that substantial investments will be required to meet renewable hydrogen demand. In general, the increase is in line with literature analysed. Still, the relative amount of increase differs between this analysis (+21%) and literature (+3 – 4%). While assumptions (3.3) are likely to be a main contributor to this difference, the investment choices for this cost increase are examined in more detail in the following chapters.

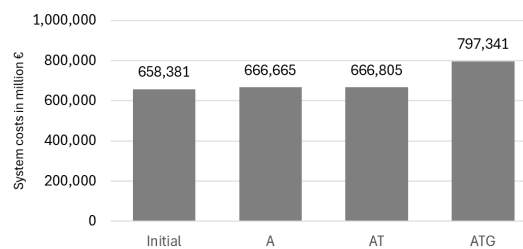


Figure 1: System costs of the scenarios [in million €]

4.2 New Capacity

The variable *NewCapacity* refers to newly built capacities in each period. Since the renewable hydrogen regulation, targets renewable energies, such as PV, offshore wind, onshore wind, run-of-river hydropower and hydropower from reservoirs, only these technologies are considered. Hydropower has little impact on the model, as existing capacities are excluded due to additionality and expansion is minimal. Still, the few new capacities are used effectively because they provide a stable supply of renewable electricity. In the base case without restrictions, the model begins building RE only in 2040, due to the availability of existing residual capacities. The total amount of newly built RE across all periods is 870.7 GW.

With the introduction of the additionality requirement, a small share of new RE shifts to 2020 and 2030 (Figure 2). This is because electrolyzers are allowed to use only newly built RE in each period, excluding residual capacities. However, the amounts of new built RE remain low since electrolyser capacities in those periods are still small. The overall total remains at 870.7 GW.

The introduction of the temporal restriction increases total new capacity of RE to 892.3 GW, with a increase observed in 2040 and 2050. This restriction requires RE capacity to be available in every modelled hour to meet electrolyser demand (Figure B 3). In 2030, new RE capacity increases by 7%, driven by a significant rise in electrolyser capacity. Since hydrogen can only be produced when new RE is available, more electrolysers are required to meet demand. In 2040 and 2050, there is a significant increase in electrolyser capacity, while the overall expansion of RE is more moderate. However, these periods see a clear shift towards the use of wind energy alongside PV (Figure 2). Wind energy provides a more stable and continuous electricity supply over the day and under changing weather conditions. This makes it a particularly effective and attractive option for powering electrolysers, despite its higher investment costs.

The geographic restriction further increases total new capacity of RE to 1123 GW, but with a major shift in periods. Electricity used by electrolysers must now be produced in the same hour and region. As a result, the model builds a large share of RE, especially onshore wind, as early as 2020 (Figure 2). In 2030, almost no new RE or electrolyser capacity is added (Figure B 2 & Figure B 3). Although onshore wind at 1100 M €/GW is more expensive than PV at 320 M €/GW, it is still much cheaper than offshore wind at 2600 M €/GW and provides better availability. Therefore, the model finds it more efficient to build more RE in 2020 and use it in 2030 and beyond. In 2050, new RE capacity increases by 85% compared to the case with only the temporal restriction, while in 2040 there is a decrease of 13% (Figure 2). Although hydrogen demand remains the same, the stricter constraint requires significantly more electrolyser capacity, which again increases the need for more RE (Figure B 2). Offshore wind plays a larger role in this period, likely due to its higher capacity factors, which enable better utilisation of electrolysers.

In summary, the step-by-step introduction of restrictions results in a 29% increase in total newly built RE capacity compared to the base case. This leads to a significant rise in costs, as seen in the objective value. The implementation of all three restrictions also shifts new RE development into earlier periods, reflecting one of the main goals of the renewable hydrogen regulation.

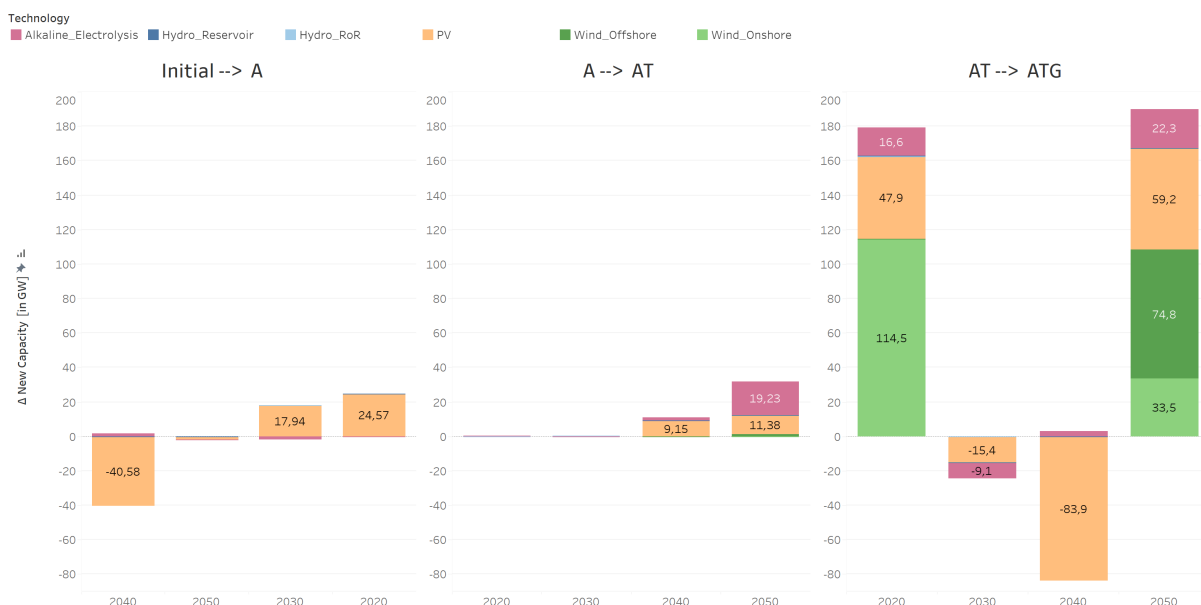


Figure 2: Differences in RE and electrolyser capacities between scenarios [in GW]

4.3 Total Capacity

The variable *TotalCapacity* represents the total available capacity per period. Since the regulatory framework for renewable hydrogen focuses only on RE, only these technologies are included in the analysis. The specific changes across the different regulatory steps and time periods were already discussed in the section on new capacities. This section focuses on how total capacities, including residual capacities, are affected in each period.

Similar to the development of new capacities, a steady increase in total capacity can be observed across the gradual introduction of restrictions (Figure 3). From the initial scenario to the final regulatory step (scenario ATG), total capacity rises by 26.4%. Additionally, capacity shifts to earlier periods, especially 2020 and 2030, although every period shows growth across all scenarios (Figure B 4).

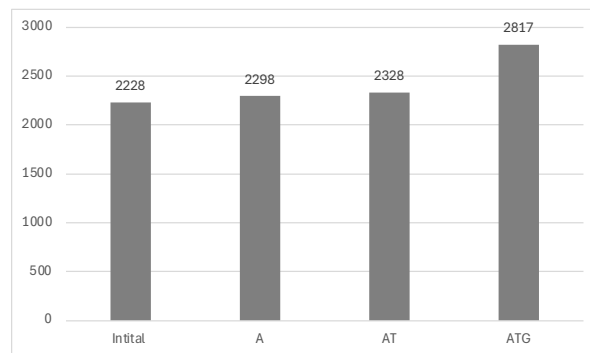


Figure 3: Development of total RE capacity across scenarios [in GW]

Electrolyser capacity also increases clearly over time and across scenarios (Figure 4). The growth over time results from rising hydrogen demand. Growth in all scenarios is driven by the stricter restrictions. These restrictions reduce the availability of electricity for electrolyzers, initially through the link to new renewable energies, then through the temporal correlation for hydrogen production, and later through the need for regional correlation. To compensate, more electrolyser capacity is needed to make better use of the available operating hours and still meet demand (Figure B 5).

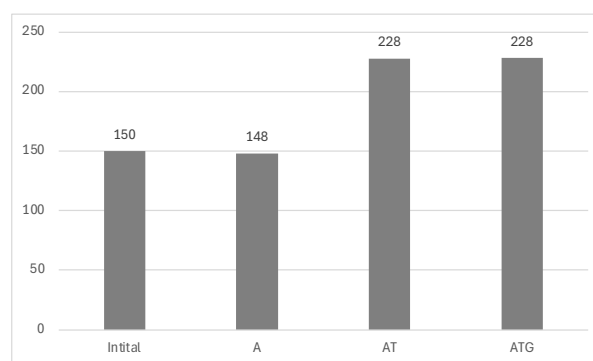


Figure 4: Development of total electrolyser capacity across scenarios [in GW]

Overall, electrolyser capacity increases by 52.2% from the initial scenario to scenario ATG, with the largest share of this increase caused by the temporal restriction. This rise in total installed capacities results in a significant increase in costs, which is reflected in the objective value.

4.4 Renewable Power Generation and Use for Hydrogen

Regulatory requirements for renewable hydrogen reshape the scale of electricity generation and shift the demand patterns of hydrogen production. The following section discusses these impacts in detail.

Figure 5 illustrates the changes in power generation and use for hydrogen that occur in the different regions with each additional level of regulation. It provides a qualitative comparison and highlights the interactions and impacts of introducing renewable hydrogen into the energy system. In Figure 5, the pink bar represents electricity consumption by electrolysis, while the other bars indicate renewable power generation. A negative value of the electrolysis bar reflects an increase in electricity consumption, whereas a positive value indicates a reduction. Conversely a positive value for the renewable bars denotes an increase in generation, while a negative signifies a decrease. Complementary absolute values for each scenario are presented in Table B 3.



Figure 5: Changes in renewable power generation and use across scenarios [GWh]

Note: There is a difference in scale between the first two diagrams and the last two.

The introduction of additionality leads to an increase in renewable electricity generation. This is explained by the fact that existing residual capacities continue to produce electricity, but cannot be used for electrolysis, making it necessary to build additional new capacity. In the model, this effect remains small, as the initial scenario already requires significant expansion due to the lack of residual capacities. Accordingly, the difference compared to additionality is minor. The electricity generation from hydro reservoirs rises marginally in France and Norway and PV electricity generation in Germany, France and the UK show a slight increase, as PV is preferred due to its comparatively low levelized cost of electricity. In addition, electricity use for electrolysis partly shifts from France to Germany and from Norway to the UK. This results from the requirement to use only renewable electricity and the costs associated with transporting electricity and hydrogen. The model therefore prioritizes the expansion of electrolyser capacities at locations that combine high renewable generation potential with direct hydrogen demand, thereby reducing transport costs.

With the introduction of temporal correlation, hydrogen may only be produced in hours when renewable electricity is available. Since electrolyzers operate at constant capacity while RE generation fluctuates strongly, two options arise. Either a significant expansion of electrolyser capacity to capture

renewable peaks or a stronger build-out of RE to supply existing electrolyzers more continuously. The model opts for the most cost-effective middle ground. The electricity generation increases in France, Norway and the UK and once again, PV expansion is preferred in the model. In the UK, some additional electricity comes from wind power. Wind power use focuses on offshore wind, as its higher full load hours provide greater electricity output and lower levelized costs than onshore wind. Some of the electrolysis capacity previously shifted to Germany is now redistributed to other countries. This is because electricity must now be produced in the same hour and PV factors in Germany are not optimal. It is therefore more cost effective for the model to allocate capacity to Spain, France, Norway and the UK, while increasing hydrogen imports from France and Denmark to Germany (Table B 4).

The introduction of geographic correlation results in the largest shifts in production and use within the model. Germany shows strongest increase in hydrogen production, accompanied by substantial expansion of renewable electricity generation. Overall, renewable electricity generation increases in the ATG scenario across all regions, underscoring the significant system impact of geographic correlation. For the first time a significant increase in wind energy use can be observed, with a marked rise in the use of onshore wind. This is mainly because offshore wind has very low maximum allowable capacity in 2020, while renewable electricity is already required from that year onward. As a result, the model prioritizes onshore wind. Once these capacities are in place, it remains more cost-effective to continue relying on them rather than switching to offshore wind. Nevertheless, Table B 3 shows that from 2050 onward, offshore wind will also be used, particularly in the UK and Germany and to a lesser extent in Spain and Norway. Due to geographic correlation hydrogen production shifts from Norway to Germany, France and the UK. This suggests that in the AT scenario a significant share of imported electricity was used for hydrogen production in Norway, but as this is no longer permitted in the ATG scenario, production is relocated to other regions.

When moving from the default scenario to the full set of regulations, renewable generation increases in all countries. The changes are driven most strongly by geographic correlation, while the other two regulations make only a minor contribute to overall system transformation. Hydrogen production increases sharply in Germany, while it declines in Norway due to the limited potential for wind and PV expansion. The effects of the different steps partly reinforce each other and partly offset one another. A clear example is the shift in hydrogen production between Germany, Norway and France, as well as the additional expansion of renewables, which increases further with each step.

In general, hydropower reservoirs show little variation between the scenarios in Spain, France and Norway, as it is non-volatile and can supply electricity continuously (Table B 3) and it can barely be extended due to the maximum capacity limitation. Hydro power run-of-river plays no role in hydrogen production, as there is little further potential for expansion, because of the maximum capacity limitation. Denmark also plays a minor role due its low hydrogen demand and the resulting low hydrogen production (Table B 3).

4.5 Renewable Share

The RE-share represents the percentage of renewable electricity generation relative to total power generation. It rises noticeably with the introduction of renewable hydrogen. While the renewable share in the initial scenario amounts 44.99% it reaches 58.4% once all regulations for renewable hydrogen are applied. This demonstrates that the regulations not only ensure climate-neutral hydrogen, but also drive additional renewable generation, making more clean electricity available for other uses. Figure B 6 shows the renewable share for each scenario as well as the total RE generation of the individual renewable technologies.

4.6 Emissions

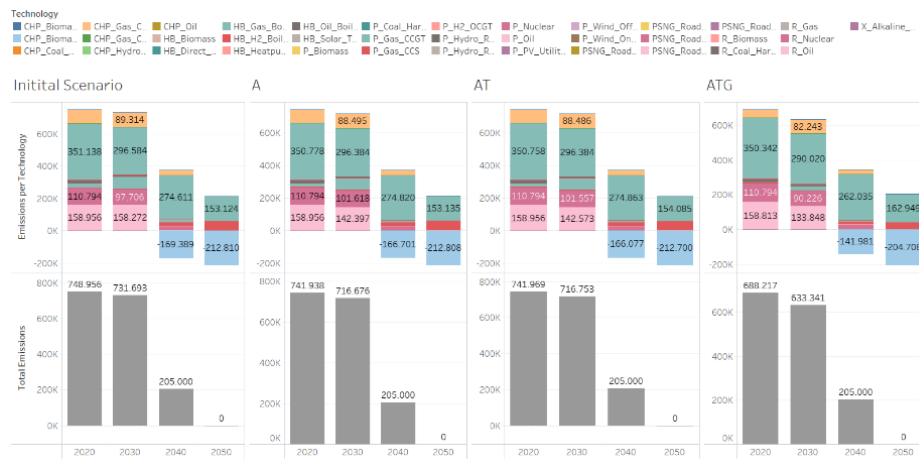


Figure 6: Emissions in different scenarios from all technologies [in tons]

The annual emissions cap underlying the model follows a gradually tightening reduction pathway, with a moderate decrease until 2030, a steep decline by 2040, and complete decarbonisation by 2050, as illustrated in Figure B 1.

The distribution of emissions in the various scenarios is shown in Figure 6. Total emissions remain well below the maximum emission limits in 2020 and 2030. It is important to note, however, that emission reductions across all scenarios do not actually occur until after 2040, remaining at similar levels beforehand. Moreover, negative emissions from carbon capture and storage only begin to play a significant role from 2040 onwards, and it is solely due to this technology that total emissions reach zero by 2050.

Throughout the entire period under consideration, heat generation from gas boilers remains the dominant source of emissions, although its absolute amount is reduced by half over time. Another important step towards achieving substantial emission reductions in the critical year 2040 is the phase-out of synthetic natural gas used in the road transport sector.

Notably, the implementation of the regulatory requirements leads to a 22.1% decrease in negative emissions from the use of biomass combined with CCS in 2040 compared to the baseline scenario. This can be explained by the fact that natural gas may no longer be used for electricity generation to produce hydrogen, instead, only renewable energy sources are permitted for hydrogen production.

4.7 Trade

Total aggregated trade data shows no changes between scenarios regarding the attribute of regions being a net import or export region. In all scenarios, the largest energy export region is Norway (initial: 2184 TWh, ATG: 2016 TWh). The largest import region is Germany (scenario: -2154 TWh, ATG: -2002 TWh). Spain is an energy export region across all scenarios (initial: 46 TWh, ATG: 88 TWh). Interestingly, it net imports hydrogen in 2050 for ATG as it is more cost efficient to import rather than produce. While trade increases for Spain due to its high PV capacity and power export, the combined trade differences reveal an overall trade decline which can be seen in Figure 7. From initial to A, gas trade slightly increases by 6.38% while hydrogen decreases by 2.62%, Power trade is nearly constant between models. From AT to A the gas trade amount decreases by 14.10% and the hydrogen trade amount increases by 1.13%, while Power decreases by 1.89%. The biggest change is the introduction of the geographical constraint, leading to a 26.62% decrease in aggregated total trade for hydrogen. If compared to the initial scenario, the ATG model has a 3.26% increase of traded gas, a 27.74% hydrogen decrease and an increase of 4.19% in traded power. This is in line with the system cost increase (4.1) as the energy system doesn't include price zones and promotes regions to meet own demand by own production (4.4). This leads to higher regional electrolyser and renewable energy capacity investments, less interregional trade and therefore higher system cost.

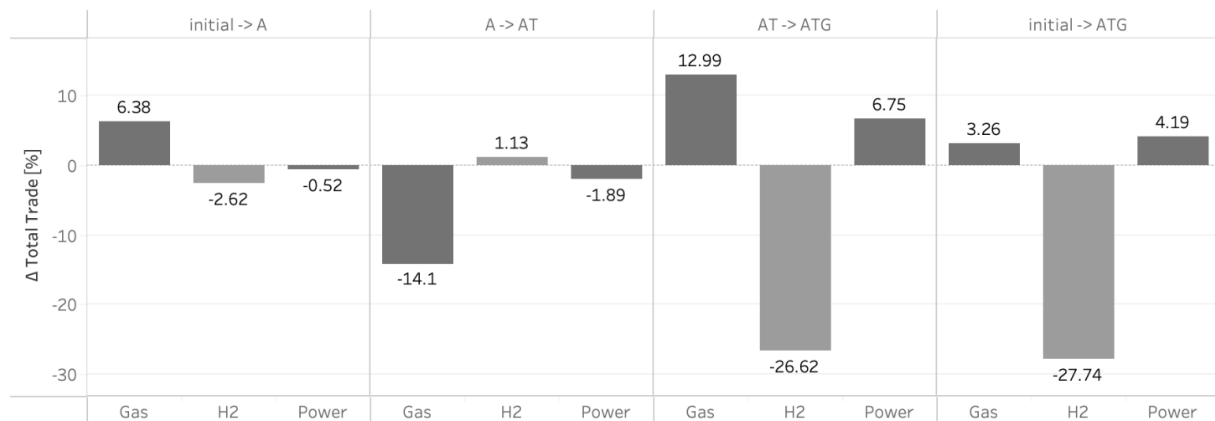


Figure 7: Total trade differences

4.8 Unmet Demand

Through analysis of the energy balance, the data does not reveal any unmet demand for the initial scenario, A and AT. This is different for the ATG model: During 2020 and 657 real hours (about 27 days) in Denmark, the demand of an aggregated amount of 39 GWh is not matched. With the underlying VoLL (20M €/GWh), this results in a total cost of unmet demand of about 781M € and accounts for 0,53% of the total system cost difference (ATG vs. initial scenario).

Through introduction of the geographic constraint, energy cost rise (4.1). In the described situation above, the marginal cost of available energy exceeds the assumed VoLL. Despite the affected monetary and energy amount being comparatively small, this indicates a risk for high price periods, lower grid stability and lower reliability because of a potential supply and demand mismatch. It is therefore necessary to examine this and similar (future) scenarios to prevent harmful system behaviour.

The use of unmet demand may be an indicator for techno-economic infeasibility of the EU regulations. As the amount highly depends on the assumed VoLL (which varies highly between sectors) and cost-overestimating limitations like the missing price zones to allow green hydrogen trade, this does not directly lead to system infeasibility. It must rather be interpreted as a risk. Possibilities to reduce marginal costs and enhancements of system flexibility must be assessed. In a real-world scenario, a strict application of renewable hydrogen constraints must be balanced with available power supply. In the event of major risks (e.g. (partial) blackouts), non-constraint electrolyser operation must be possible.

5 Conclusion

This term paper examined the effects of renewable hydrogen regulations on an energy system model by stepwise implementing additionality, temporal correlation, and geographic correlation. The results show that these regulations significantly impact system costs, renewable energy deployment, electrolyser capacity.

System costs increased by 21.11% in the fully regulated scenario (ATG), with geographic correlation causing the largest rise. Newly built renewable capacity grew by 29%, and total system capacity increased by 26.4%, driven by earlier and stronger investments in wind energy, especially in 2040 and 2050. Electrolyser capacity expanded by 52.2%, reflecting the need to meet stricter conditions on renewable electricity use.

The regulations led to major shifts in renewable electricity generation and hydrogen production. PV and onshore wind were preferred due to cost and availability, and hydrogen production increasingly concentrated in regions with strong renewable potential. While the renewable electricity share increased from 45% to 58.4%, total emissions remained stable until 2040 and reached net-zero by 2050 through CCS. Trade volumes decreased as the system relied more on regional production, and unmet demand occurred only in the ATG scenario, indicating potential risks under strict enforcement.

Overall, the analysis shows that renewable hydrogen regulations support decarbonisation and the expansion of renewables but also increase system costs and introduce operational risks. However, these findings must be interpreted with caution, as the study relies on several simplifying assumptions that improve tractability but limit external validity. An independent hydrogen supply is not possible given the actual data and constraints. The inclusion of Spain in the model did not change it fundamentally.

The model uses 120 representative hours per year and decadal decision steps (2020–2050). Hourly temporal correlation is enforced earlier than required in practice, and transitional provisions (e.g., monthly matching until 2029, 36-month additionality window) are not applied. Geographic constraints are implemented at the country level rather than by bidding zone. Offshore bidding zones and price-based cross-zonal eligibility are not represented. Power from storage is not allowed to be used in electrolyzers. This reduces modelled cross-zonal flexibility and tends to overstate costs and domestic capacity needs. These likely overestimates system costs and renewable overbuild.

Future work should lift the temporal resolution to full-year hourly data and explicitly implement the RED III transition phases. By implementing resource constraints for water availability and modelling of energy flows the traceability can be increased.

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Appendix A

Julia code:

```

255 # insert the unmetdemandpenalty in the objective function
256 # the objective function
257 # total costs should be minimized
258 @objective(ESM, Min,
259     sum(TotalCost[y,r,t]/ (1+DiscountRate)^(y - minimum(year)) for t in technologies for r in regions, y in year)
260     + sum(TotalStorageCost[y,r,s]/ (1+DiscountRate)^(y - minimum(year)) for s in storages for r in regions, y in year)
261     + sum(TradeCostFactor[f]*TradeDistance[r,rr] * Export[y,rr,r,h,f] * YearlyDifferenceMultiplier[y] / (1+DiscountRate)^(y - minimum(year))
262     for h in hour, r in regions, rr in regions, f in fuels, y in year if MaxTradeCapacity[y,r,rr,f]>0)
263     - sum(SalvageValue[y,r,t] for y in year, r in regions, t in technologies)
264     + sum(UnmetDemand[y,r,h,f]*UnmetDemandPenalty[y,r] for y in year, r in regions, h in hour, f in fuels)
265 )

```

Figure A 1: New Objective Function: Code from the file 00_default.jl

```

41 # We need to read in our new input parameters via new csv files
42 TagRenewable = readin("tagrenewable.csv",default=0, dims=1, dir=data_dir)
43 TagElectrolyzer = readin("tagelectrolyzer.csv", default=0, dims=1, dir=data_dir)
44
45 # we need to read in region and year dependent unmetdemandpenalty
46 UnmetDemandPenalty = readin("unmetdemandpenalty.csv", default=9999, dims=2, dir=data_dir)

```

Figure A 2: New Parameter: Code from the file 01_additionality.jl

```

260 ##### new constraints for term paper #####
261
262 #only renewable power is allowed to use for hydrogen production
263 #renewable power production greater than or equal electrolyser power use
264 #y in year, f in fuels as indices
265 @constraint(ESM, RenewablePowerForHydrogen[y in year, f in fuels; f == "Power"],
266     sum(FuelProductionByTechnology[y,r,h,t,f]
267         for r in regions, h in hour
268         for t in technologies if TagRenewable[t] == 1) >=
269     sum(FuelUseByTechnology[y,r,h,t,f]
270         for r in regions, h in hour
271         for t in technologies if TagElectrolyzer[t] == 1)
272 )
273
274 #power input in new electrolyser capacities coupling with power output of new renewable capacities
275 #y in year, f in fuels as indices
276 @constraint(ESM, NewCapacityCoupling[y in year, f in fuels; f == "Power"],
277     sum(NewCapacity[y,r,t] * InputRatio[t,f]
278         for r in regions, t in technologies if TagElectrolyzer[t] == 1) <=
279     sum(NewCapacity[y,r,t] * OutputRatio[t,f]
280         for r in regions, t in technologies if TagRenewable[t] == 1)
281 )
282
283
284 #Restriction of the annual electricity consumption of the new electrolysis capacities by the new capacities available this year
285 #y in year, f in fuels as indices
286 @constraint(ESM, EqualPowerUseAndProduction[y in year, f in fuels; f == "Power"],
287     #Electricity generation of new renewable capacities
288     sum(NewCapacity[y,r,t] * CapacityFactor[r,h,t] * OutputRatio[t,f] * 8760/n_hour
289         for r in regions
290         for h in hour
291         for t in technologies if TagRenewable[t] == 1) >=
292     #Electricity consumption of all electrolysis capacities
293     sum(FuelUseByTechnology[y,r,h,t,f]
294         #Electricity consumption of the old electrolysis capacities
295         - InputRatio[t,f] * (AccumulatedCapacity[y,r,t] - NewCapacity[y,r,t]) * CapacityFactor[r,h,t] * 8760/n_hour
296         for r in regions
297         for h in hour
298         for t in technologies if TagElectrolyzer[t] == 1)
299 )
300 )

```

Figure A 3: Additionality Constraints: Code from the file 01_additionality.jl

```

262 ##### new constraints for term paper #####
263
264 #only renewable power is allowed to use for hydrogen production
265 #renewable power production greater than or equal electrolyser power use
266 #y in year, h in hour, f in fuels as indices
267 @constraint(ESM, RenewablePowerForHydrogen[y in year, h in hour, f in fuels; f == "Power"],
268     sum(FuelProductionByTechnology[y,r,h,t,f]
269         for r in regions
270         for t in technologies if TagRenewable[t] == 1) >=
271     sum(FuelUseByTechnology[y,r,h,t,f]
272         for r in regions
273         for t in technologies if TagElectrolyzer[t] == 1)
274 )
275
276 #power input in new electrolyser capacities coupling with power output of new renewable capacities
277 #y in year, f in fuels as indices
278 @constraint(ESM, NewCapacityCoupling[y in year, f in fuels; f == "Power"],
279     sum(NewCapacity[y,r,t] * InputRatio[t,f]
280         for r in regions, t in technologies if TagElectrolyzer[t] == 1) <=
281     sum(NewCapacity[y,r,t] * OutputRatio[t,f]
282         for r in regions, t in technologies if TagRenewable[t] == 1)
283 )
284
285 #Restriction of the annual electricity consumption of the new electrolysis capacities by the new capacities available this year
286 #y in year, h in hour, f in fuels as indices
287 @constraint(ESM, EqualPowerUseAndProduction[y in year, h in hour, f in fuels; f == "Power"],
288     #Electricity generation of new renewable capacities
289     sum(NewCapacity[y,r,t] * CapacityFactor[r,h,t] * OutputRatio[t,f] * 8760/n_hour
290         for r in regions
291         for t in technologies if TagRenewable[t] == 1) >=
292     #Electricity consumption of all electrolysis capacities
293     sum(FuelUseByTechnology[y,r,h,t,f]
294     #Electricity consumption of the old electrolysis capacities
295     - InputRatio[t,f] * (AccumulatedCapacity[y,r,t] - NewCapacity[y,r,t]) * CapacityFactor[r,h,t] * 8760/n_hour
296         for r in regions
297         for t in technologies if TagElectrolyzer[t] == 1)
298 )
299 )

```

Figure A 4: Temporal Constraints: Code from the file 02_temporal.jl

```

258 ##### new constraints for term paper #####
259
260 #only renewable power is allowed to use for hydrogen production
261 #renewable power production greater than or equal electrolyser power use
262 #y in year, h in hour, r in regions, f in fuels as indices
263 @constraint(ESM, RenewablePowerForHydrogen[y in year, h in hour, r in regions, f in fuels; f == "Power"],
264     sum(FuelProductionByTechnology[y,r,h,t,f]
265         for t in technologies if TagRenewable[t] == 1) >=
266     sum(FuelUseByTechnology[y,r,h,t,f]
267         for t in technologies if TagElectrolyzer[t] == 1)
268 )
269
270 #power input in new electrolyser capacities coupling with power output of new renewable capacities
271 #y in year, r in regions, f in fuels as indices
272 @constraint(ESM, NewCapacityCoupling[y in year, r in regions, f in fuels; f == "Power"],
273     sum(NewCapacity[y,r,t] * InputRatio[t,f]
274         for t in technologies if TagElectrolyzer[t] == 1) <=
275     sum(NewCapacity[y,r,t] * OutputRatio[t,f]
276         for t in technologies if TagRenewable[t] == 1)
277 )
278
279 #Restriction of the annual electricity consumption of the new electrolysis capacities by the new capacities available this year
280 #y in year, h in hour, r in regions, f in fuels as indices
281 @constraint(ESM, EqualPowerUseAndProduction[y in year, h in hour, r in regions, f in fuels; f == "Power"],
282     #Electricity generation of new renewable capacities
283     sum(NewCapacity[y,r,t] * CapacityFactor[r,h,t] * OutputRatio[t,f] * 8760/n_hour
284         for t in technologies if TagRenewable[t] == 1) >=
285     #Electricity consumption of all electrolysis capacities
286     sum(FuelUseByTechnology[y,r,h,t,f]
287     #Electricity consumption of the old electrolysis capacities
288     - InputRatio[t,f] * (AccumulatedCapacity[y,r,t] - NewCapacity[y,r,t]) * CapacityFactor[r,h,t]
289     for t in technologies if TagElectrolyzer[t] == 1)* 8760/n_hour
290 )
291 )

```

Figure A 5: Geographical Constraints: Code from the file 03_temporal.jl

Appendix B

Region	VoLL Range for years 2020 – 2050 [M €/ GWh]	Reference
DE	12 - 18	(Röpke 2013; Piaszeck et al. 2013)
DK	20 - 26	(Energienet 2022)
FR	18 - 24	(Capacity Remuneration Mechanism 2020)(Belgium values)
NO	5 - 6,5	(Kjolle et al. 2008)
UK	23 - 30	(London Economics 2013)
ES	10 – 13	(Linares and Rey 2013)

Table B 1: Values of Lost Load for different regions and years

Year	Trade Region	Fuel	Value [GW]
2020	France	Gas	7,0
2020	France	Power	2,8
2020	France	Hydrogen	0,0
2030	France	Gas	9,3
2030	France	Power	5,0
2030	France	Hydrogen	7,5
2040	France	Gas	9,3
2040	France	Power	9,5
2040	France	Hydrogen	12,0
2050	France	Gas	9,3
2050	France	Power	9,5
2050	France	Hydrogen	30,0
2020 - 2050	All remaining	all	0,0

Table B 2: Trade capacity data for Spain

Data: Gas (Enagás 2023), Power: (ENTSO-E 2024), Hydrogen: (ENTSOG AISBL 2025))

Initial scenario

	DE				DK				ES				FR				NO				UK			
Technology	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050
Alkaline_Electrolysis	-102.114	-40.716	-114.661	-287.289	-1.751	-152	-569	0	-35.302	-37.025	-47.371	-69.804	-33.552	-139.341	-196.394	-137.284	-9.920	-81.140	-87.808	-99.395	-34.427	-4.848	-8.357	-116.449
Hydro_Reservoir	12.614	12.614	12.614	12.614	0	0	0	0	167.535	167.579	167.579	167.579	75.143	76.975	76.975	76.975	185.926	233.550	233.550	233.550	0	0	0	0
Hydro_RoR	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PV	38.679	38.679	233.134	233.134	879	838	20.175	20.175	8.352	7.349	130.831	130.831	7.215	8.956	146.447	298.867	0	0	0	0	10.233	10.558	123.178	151.945
Wind_Offshore	0	0	0	0	0	0	0	0	0	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind_Onshore	113.538	114.931	0	0	9.167	8.615	0	0	46.803	46.076	0	0	24.797	24.384	0	0	0	0	0	0	29.857	13.295	0	0

scenario A

	DE				DK				ES				FR				NO				UK			
Technology	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050
Alkaline_Electrolysis	-102.114	-56.414	-113.707	-287.378	-1.751	-322	-568	0	-35.302	-37.025	-47.371	-69.813	-33.552	-121.526	-197.581	-137.278	-9.920	-80.694	-87.626	-99.294	-34.427	-5.554	-8.404	-116.454
Hydro_Reservoir	12.614	12.614	12.614	12.614	0	0	0	0	167.579	167.579	167.579	167.579	76.975	76.975	76.975	76.975	187.657	233.550	233.550	233.550	0	0	0	0
Hydro_RoR	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PV	54.063	70.276	233.134	233.134	879	838	20.175	20.175	8.352	7.349	130.831	130.831	7.215	8.956	147.895	298.867	0	0	0	0	16.374	16.699	123.577	152.021
Wind_Offshore	0	0	0	0	0	0	0	0	0	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind_Onshore	113.538	114.931	0	0	9.167	8.615	0	0	46.803	46.076	0	0	24.797	24.384	0	0	0	0	0	0	29.857	13.295	0	0

scenario AT

	DE				DK				ES				FR				NO				UK			
Technology	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050
Alkaline_Electrolysis	-102.114	-56.473	-111.464	-285.729	-1.751	-323	-452	0	-35.302	-37.025	-47.094	-71.714	-33.552	-121.433	-198.665	-137.108	-9.920	-80.721	-87.406	-103.201	-34.427	-5.553	-9.983	-117.270
Hydro_Reservoir	12.614	12.614	12.614	12.614	0	0	0	0	167.579	167.579	167.579	167.579	76.975	76.975	76.975	76.975	187.645	233.550	233.550	233.550	0	0	0	0
Hydro_RoR	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PV	53.886	70.206	233.134	233.134	879	838	20.175	20.175	8.352	7.349	130.831	130.831	7.215	8.956	155.884	298.867	0	0	0	6.145	16.432	16.757	124.561	162.092
Wind_Offshore	0	0	0	0	0	0	0	0	0	19	0	0	0	0	0	0	0	0	0	0	0	0	0	4.498
Wind_Onshore	113.538	114.931	0	0	9.167	8.615	0	0	46.803	46.076	0	0	24.797	24.384	0	0	0	0	0	0	29.857	13.295	0	0

scenario ATG

	DE				DK				ES				FR				NO				UK			
Technology	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050	2020	2030	2040	2050
Alkaline_Electrolysis	-104.884	-96.596	-144.579	-280.526	-1.682	-275	-1.262	0	-36.480	-46.683	-58.752	-50.657	-34.780	-90.738	-150.759	-240.580	-10.105	-49.153	-70.730	-34.101	-34.666	-16.936	-29.402	-113.578
Hydro_Reservoir	12.614	12.614	12.614	12.614	0	0	0	0	165.284	167.579	167.579	167.579	76.975	76.975	76.975	76.975	178.319	229.598	233.550	228.957	0	0	0	0
Hydro_RoR	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PV	69.564	69.564	233.134	233.134	1.044	1.003	20.175	20.175	28.948	27.945	110.937	130.831	26.545	28.286	135.577	298.867	1.677	3.699	3.778	12.048	10.233	10.558	108.141	162.092
Wind_Offshore	0	0	0	202.685	0	0	0	0	19	0	0	13.359	0	0	0	0	0	0	0	1.906	0	0	0	91.887
Wind_Onshore	194.135	195.528	80.597	72.728	18.242	17.690	9.076	0	65.921	65.195	19.118	0	38.949	38.536	14.152	0	5.379	5.379	5.379	0	148.782	132.219	118.925	0

Table B 3: Total Renewable Power Generation and Use [in GWh]

Initial scenario --> scenario A

To	From	
DE	DK	107.846
	FR	1.251.526
DK	DE	1.225.658
	NO	3.801.083
ES	FR	1.373.182
FR	DE	1.225.658
	ES	46.119
	UK	9.310
NO	DK	344.115
	UK	9.310
UK	FR	1.241.636
	NO	4.056.132

scenario A --> scenario AT

To	From	
DE	DK	-104.320
	FR	-1.268.019
DK	DE	-1.225.658
	NO	-3.797.465
ES	FR	-1.373.182
FR	DE	-1.225.658
	ES	-45.608
	UK	-9.310
NO	DK	-344.115
	UK	-9.310
UK	FR	-1.244.487
	NO	-4.055.911

scenario AT --> scenario ATG

To	From	
DE	DK	-239.795
	FR	-105.163
DK	DE	0
	NO	-261.616
ES	FR	0
FR	DE	0
	ES	-1.632
	UK	0
NO	DK	0
	UK	0
UK	FR	-128.695
	NO	-3.171

Table B 4: Changes in H2 trade between the scenarios [in GWh]

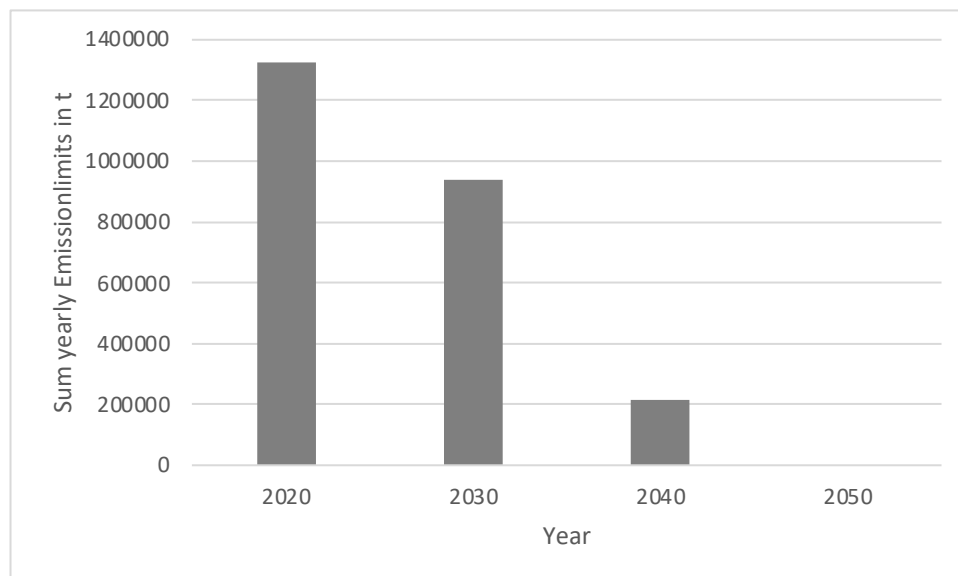


Figure B 1: Annual emission limit of the model (own representation)

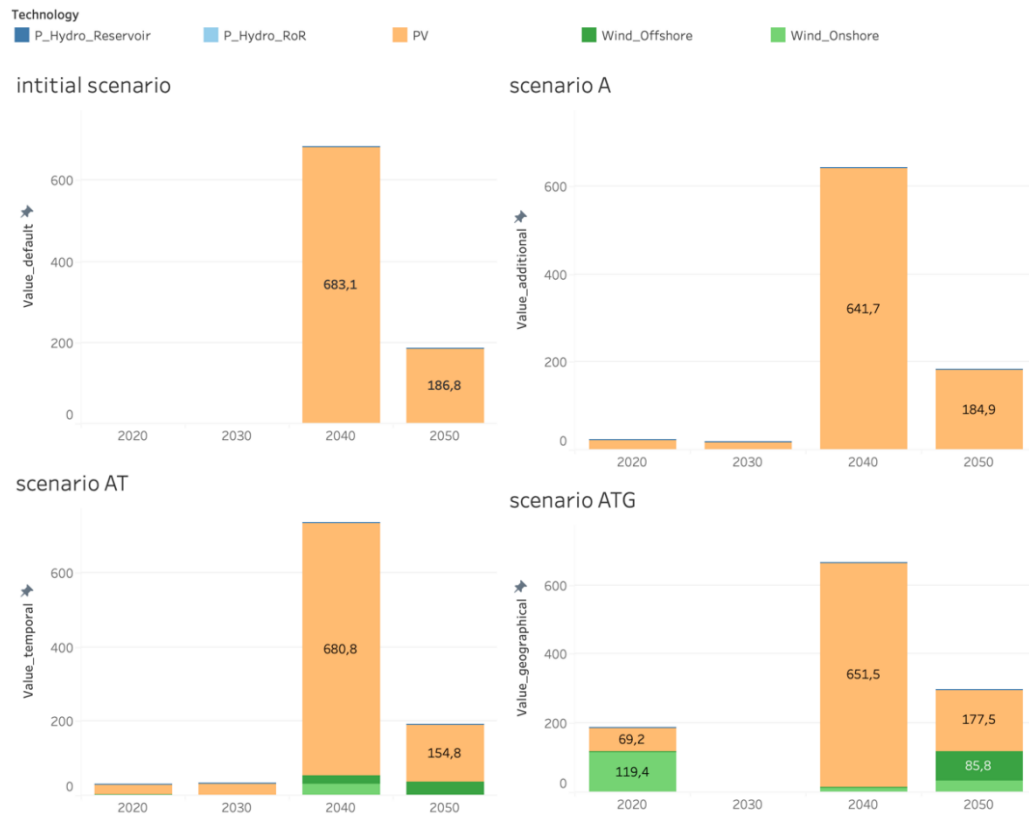


Figure B 2: New capacity of RE in GW across the different scenarios

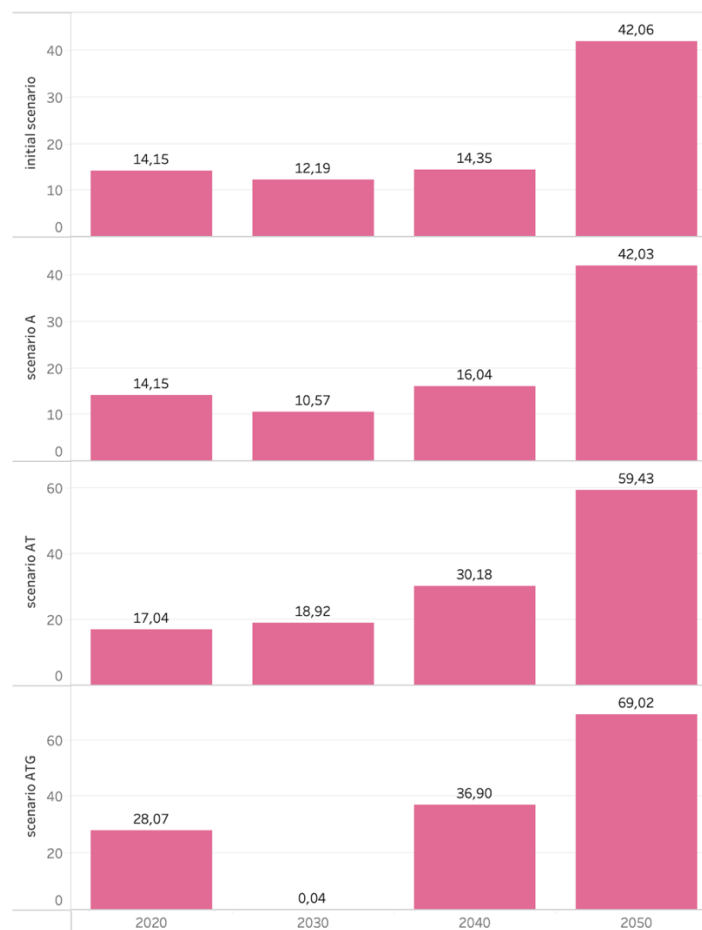


Figure B 3: New capacity of electrolysis in GW across the different scenarios

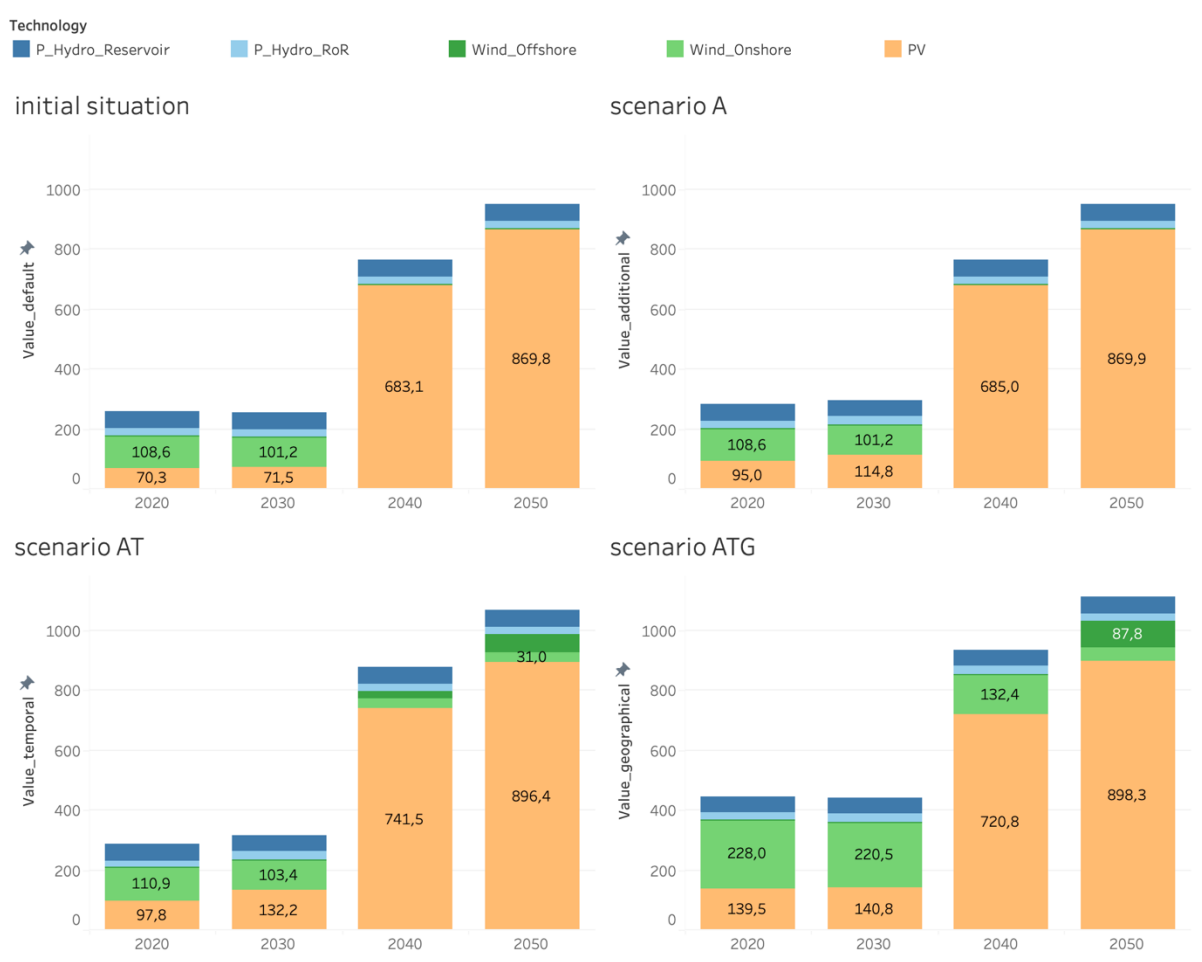


Figure B 4: Total capacity of RE in GW across the different scenarios

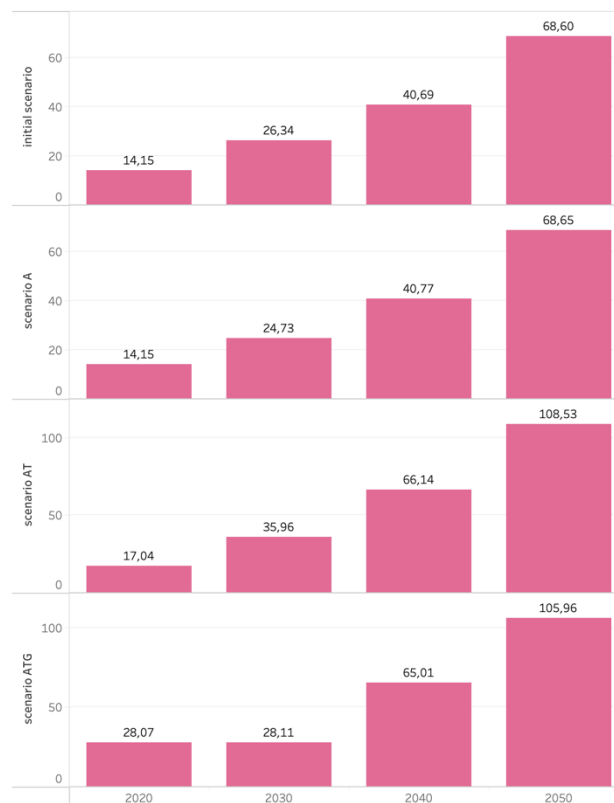


Figure B 5: Total capacity of electrolysis in GW across the different scenarios

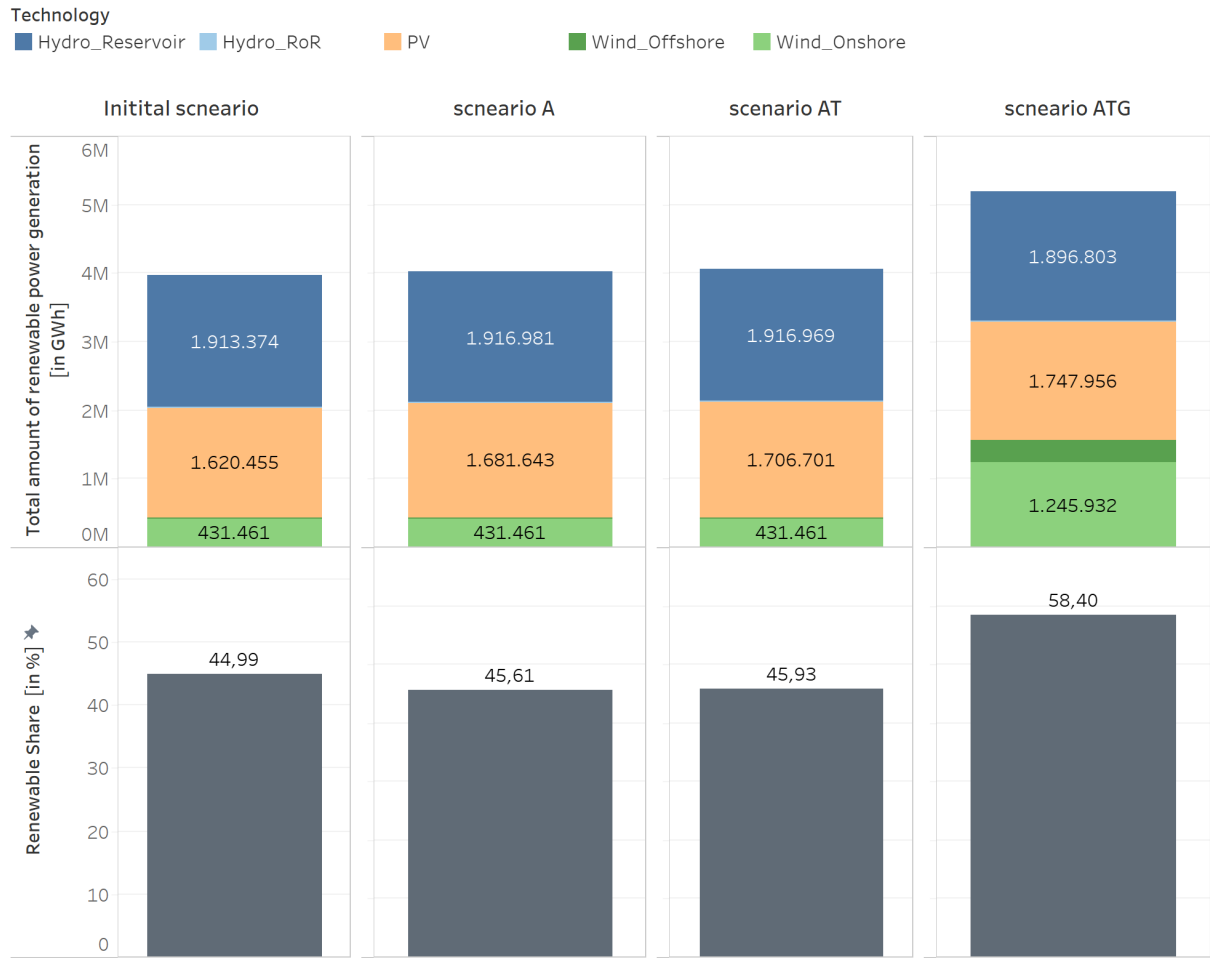


Figure B 6: Renewable Share and total amount of renewable power generation